NexHire AI – Entity Relationship Diagram

Smart Employee Recruitment & AI Hiring System · MySQL relational schema · Crow's foot notation

LEGEND

——||	Exactly one (mandatory)	——o|	Zero or one (optional)
——|<	One or many (mandatory)	——o<	Zero or many (optional)
PK	Primary key (bold, underlined)	FK	Foreign key (italic)
UK	Unique key	PK,FK	Composite key part

Users & Access

CALENDAR_CONNECTION

PK	**connection_id**	INT
FK,UK	*user_id*	INT
	google_account	VARCHAR(150)
	access_token_enc	TEXT
	refresh_token_enc	TEXT
	token_expiry	DATETIME

PASSWORD_RESET

PK	**reset_id**	INT
FK	*user_id*	INT
UK	token_hash	VARCHAR(255)
	expires_at	DATETIME
	used_at	DATETIME

USER

PK	**user_id**	INT
	full_name	VARCHAR(100)
UK	email	VARCHAR(150)
	password_hash	VARCHAR(255)
	role	ENUM(HR_ADMIN, INTERVIEWER, CANDIDATE)
	status	ENUM(ACTIVE, DEACTIVATED)
	created_at	DATETIME
	last_login_at	DATETIME

CANDIDATE_PROFILE

PK	**profile_id**	INT
FK,UK	*user_id*	INT
	phone	VARCHAR(20)
	address	VARCHAR(255)
	city	VARCHAR(60)
	highest_qualification	VARCHAR(100)
	updated_at	DATETIME

Offers, Referrals & Candidate Support

FAQ_ENTRY

PK	**faq_id**	INT
FK	*updated_by*	INT
	question	TEXT
	answer	TEXT
	category	VARCHAR(60)
	updated_at	DATETIME

OFFER

PK	**offer_id**	INT
FK,UK	*application_id*	INT
FK	*approved_by*	INT
	salary_band	VARCHAR(50)
	start_date	DATE
	role_terms	TEXT
	letter_body	TEXT
	status	ENUM(DRAFT, APPROVED, SENT)
	sent_at	DATETIME

CHAT_SESSION

PK	**session_id**	INT
FK	*candidate_id*	INT
	started_at	DATETIME

CHAT_MESSAGE

PK	**message_id**	INT
FK	*session_id*	INT
	sender	ENUM(CANDIDATE, BOT)
	message_text	TEXT
	sent_at	DATETIME

REFERRAL

PK	**referral_id**	INT
FK,UK	*application_id*	INT
FK	*marked_by*	INT
	employee_name	VARCHAR(100)
	employee_department	VARCHAR(100)
	employee_position	VARCHAR(100)
	decision	ENUM(PENDING, ACCEPTED, REJECTED)
	decided_at	DATETIME

Notifications, Integrations & Audit

AUDIT_LOG

PK	**log_id**	INT
FK	*user_id*	INT
	entity_type	VARCHAR(50)
	entity_id	INT
	action	VARCHAR(80)
	details	TEXT
	created_at	DATETIME

EMAIL_LOG

PK	**email_id**	INT
FK	*recipient_id*	INT
FK	*application_id*	INT
	email_type	ENUM(CONFIRMATION, SHORTLIST, REJECTION, INTERVIEW, OFFER, PASSWORD_RESET)
	subject	VARCHAR(200)
	status	ENUM(QUEUED, SENT, FAILED)
	retry_count	INT
	sent_at	DATETIME

Applications & AI Screening

STAGE_HISTORY

PK	**history_id**	INT
FK	*application_id*	INT
FK	*changed_by*	INT
	from_stage	VARCHAR(20)
	to_stage	VARCHAR(20)
	changed_at	DATETIME

APPLICATION

PK	**application_id**	INT
FK	*vacancy_id*	INT
FK	*candidate_id*	INT
	portfolio_link	VARCHAR(255)
	stage	ENUM(APPLIED, SHORTLISTED, INTERVIEW, OFFER, HIRED, REJECTED)
	submitted_at	DATETIME
	updated_at	DATETIME

DUPLICATE_FLAG

PK	**flag_id**	INT
FK	*application_id*	INT
FK	*matched_application_id*	INT
	match_reason	VARCHAR(100)
	similarity	DECIMAL(5,2)
	status	ENUM(OPEN, MERGED, DISCARDED, DISMISSED)
FK	*resolved_by*	INT
	resolved_at	DATETIME

APPLICATION_ANSWER

PK	**answer_id**	INT
FK	*application_id*	INT
FK	*field_id*	INT
	answer_value	TEXT

AI_SCORE

PK	**score_id**	INT
FK,UK	*application_id*	INT
	fulfilment_pct	DECIMAL(5,2)
	rank_no	INT
	match_tag	ENUM(STRONG, POSSIBLE, NOT_A_FIT)
	predicted_score	DECIMAL(5,2)
	eval_status	ENUM(PENDING, EVALUATED, FAILED)
	tag_overridden	BOOLEAN
	evaluated_at	DATETIME

SCORE_DETAIL

PK	**detail_id**	INT
FK	*score_id*	INT
FK	*requirement_id*	INT
	is_met	BOOLEAN
	matched_value	VARCHAR(100)
	weight_awarded	DECIMAL(5,2)

Interviews & Feedback

AI_INSIGHT

PK	**insight_id**	INT
FK	*application_id*	INT
	strengths	TEXT
	concerns	TEXT
	summary	TEXT
	status	ENUM(PENDING, READY)
	generated_at	DATETIME

INTERVIEW_FEEDBACK

PK	**feedback_id**	INT
FK	*interview_id*	INT
FK	*author_id*	INT
	notes	TEXT
	created_at	DATETIME

INTERVIEW_PANEL

PK,FK	**interview_id**	INT
PK,FK	**interviewer_id**	INT
	assigned_at	DATETIME

INTERVIEW

PK	**interview_id**	INT
FK	*application_id*	INT
FK,UK	*slot_id*	INT
FK	*scheduled_by*	INT
	round_no	INT
	status	ENUM(SCHEDULED, CONDUCTED, NO_SHOW, POSTPONED, CANCELLED)
	google_event_id	VARCHAR(150)
	created_at	DATETIME

INTERVIEW_SLOT

PK	**slot_id**	INT
FK	*vacancy_id*	INT
	slot_date	DATE
	start_time	TIME
	end_time	TIME
	venue	VARCHAR(150)
	is_booked	BOOLEAN

STRUCTURED_RATING

PK	**rating_id**	INT
FK	*interview_id*	INT
FK	*interviewer_id*	INT
FK	*criterion_id*	INT
	rating	TINYINT (1-5)
	comment	VARCHAR(255)
	submitted_at	DATETIME

FEEDBACK_CRITERION

PK	**criterion_id**	INT
UK	name	VARCHAR(80)
	description	VARCHAR(255)

Vacancies & Application Forms

DEPARTMENT

PK	**department_id**	INT
UK	name	VARCHAR(100)
	description	VARCHAR(255)

VACANCY

PK	**vacancy_id**	INT
FK	*department_id*	INT
FK	*created_by*	INT
	title	VARCHAR(150)
	description	TEXT
	location	VARCHAR(100)
	status	ENUM(DRAFT, OPEN, CLOSED, FILLED)
	published_at	DATETIME
	closed_at	DATETIME

INTERVIEW_QUESTION

PK	**question_id**	INT
FK	*vacancy_id*	INT
FK	*approved_by*	INT
	question_text	TEXT
	category	ENUM(TECHNICAL, BEHAVIOURAL)
	is_approved	BOOLEAN
	created_at	DATETIME

FORM_FIELD

PK	**field_id**	INT
FK	*vacancy_id*	INT
FK	*requirement_id*	INT
	label	VARCHAR(150)
	field_type	ENUM(TEXT, NUMBER, DATE, LIST, URL)
	is_required	BOOLEAN
	display_order	INT

VACANCY_REQUIREMENT

PK	**requirement_id**	INT
FK	*vacancy_id*	INT
	req_type	ENUM(SKILL, EXPERIENCE, QUALIFICATION, CERTIFICATION)
	name	VARCHAR(100)
	min_value	VARCHAR(50)
	is_mandatory	BOOLEAN
	weightage	DECIMAL(5,2)

JOB_POSTING

PK	**posting_id**	INT
FK	*vacancy_id*	INT
	job_board	ENUM(LINKEDIN, INDEED)
	external_posting_id	VARCHAR(100)
	status	ENUM(POSTED, FAILED)
	posted_at	DATETIME